LAB EXPERIMENT - 8

EXPERIMENT 8: THREAT INTELLIGENCE COLLECTION FROM OTX & VIRUSTOTAL

TITLE: THREAT INTELLIGENCE REPORT – SOLARWINDS SUPPLY CHAIN ATTACK

➢ Aim

To analyze a real-world cyber attack and prepare a detailed threat intelligence report.

➢ Introduction

The SolarWinds Supply Chain Attack was a highly sophisticated cyber attack discovered in 2020. Attackers compromised the software update mechanism of SolarWinds, affecting thousands of organizations globally.

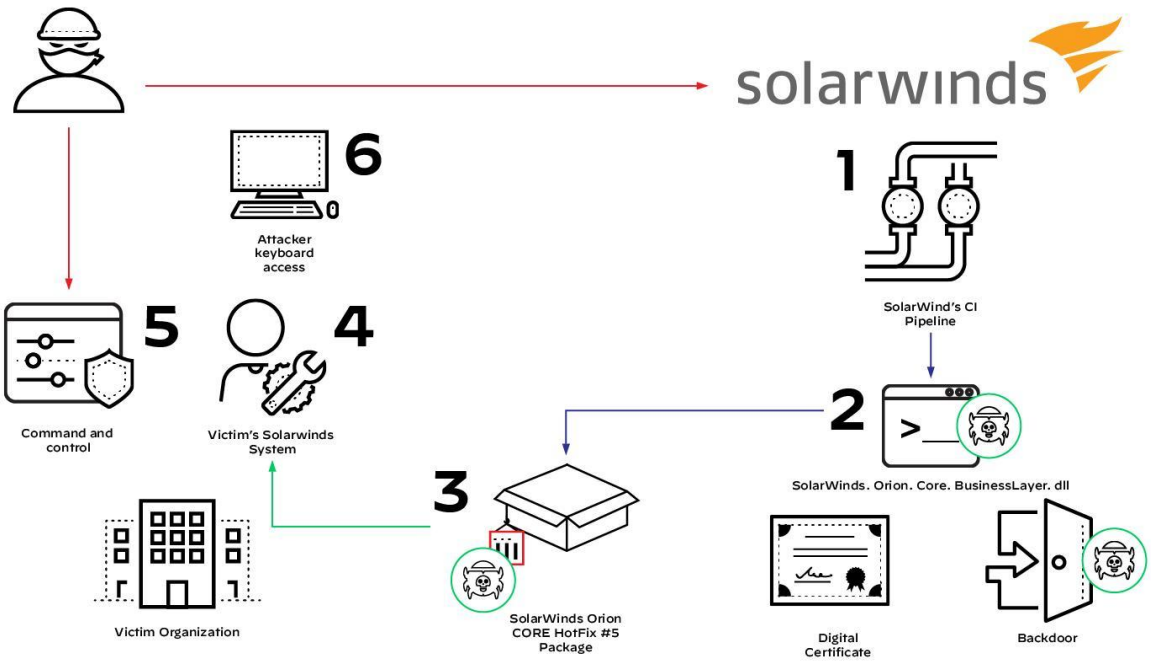
➢ Threat Type

- Supply Chain Attack
- Advanced Persistent Threat (APT)

➢ Description of the Incident

Attackers inserted malicious code into the Orion software updates of SolarWinds. When organizations installed the update, the malware was also installed, giving attackers unauthorized access.

- **Diagram Flow-**



- **Attack Methodology**

1. Attackers gained access to SolarWinds' development environment
2. Injected malicious code into software updates
3. Distributed infected updates to customers
4. Established backdoor access in victim systems

- **Indicators of Compromise (IoC)**

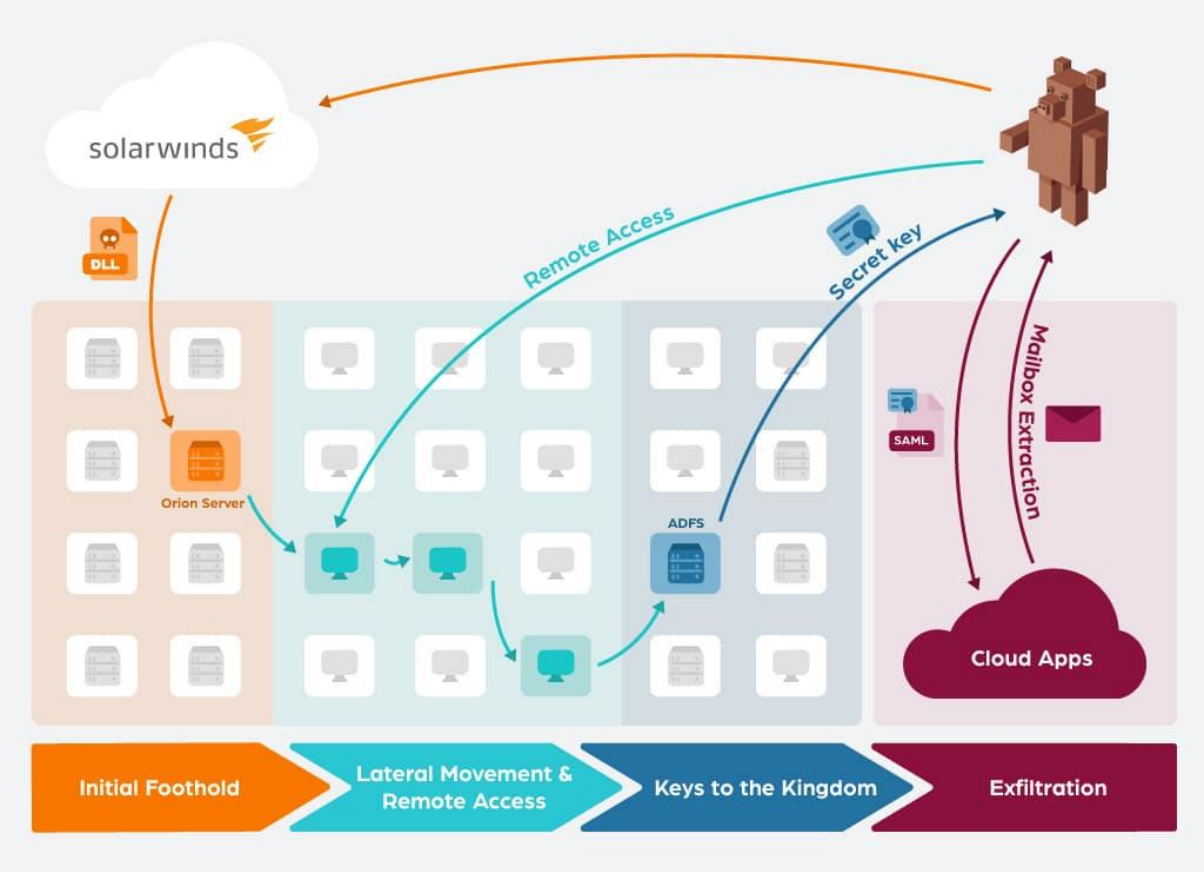
- Unexpected outbound traffic to unknown servers
- Unauthorized access to sensitive data
- Suspicious system processes
- Unusual login activity

- **Impact of the Attack**

- Affected over 18,000 organizations
- Data breaches in government and private sectors

- Long-term espionage and data theft
- Compromise of sensitive information

➢ **SolarWinds Supply Chain Attack Lifecycle-**



➢ **Attack Analysis**

The attack was highly stealthy and remained undetected for months. It used trusted software updates as a delivery mechanism, making it difficult to identify.

➢ **Mitigation Strategies**

- Verify software updates using digital signatures
- Implement zero trust security model
- Monitor network traffic continuously
- Apply timely security patches
- Use multi-factor authentication (MFA)

- **Threat Intelligence Insights**
  - Attackers targeted the supply chain instead of direct systems
  - Trust relationships can be exploited
  - Continuous monitoring is essential

- **Result**

The SolarWinds attack was successfully analyzed, and its attack vector, impact, and mitigation strategies were identified.

- **Conclusion**

Threat intelligence improves proactive defense by helping organizations detect advanced threats early and strengthen their security posture.